

Manish Kumar

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EDUCATION

- Indian Institute of Technology, Mandi** Himachal Pradesh, India
B.Tech. in Electrical Engineering; CGPA: 8.07/10.00 *August 2023 – May 2027*
 - Relevant Coursework:** Data Structures and Algorithms, Introduction to Data Science, Probability and Statistics, Machine Learning, Deep Learning and Applications, Signals and Systems

EXPERIENCE

- Zangoh (Newzera Tech Labs Pvt. Ltd.)** Indore, India
Software Engineering Intern – Generative AI *January 2026 – July 2026*
 - Developed **Python/FastAPI** backend services and **REST APIs** for **Generative AI** applications, implementing **Pydantic** validation, asynchronous workflows, and modular service components.
 - Engineered speech and lip-sync data pipelines integrating audio ingestion, shot segmentation, **MediaPipe** face and landmark extraction, **Whisper ASR**, phoneme extraction, metadata generation, and automated quality filtering.
 - Contributed to **LLM-powered AI Avatar** and **multi-agent systems** using **LangChain**, **prompt engineering**, and agent orchestration for context-aware conversational and content-generation workflows.
 - Worked with **PostgreSQL**, **Alembic**, **Docker**, **CloudEvents**, and **event-driven microservices**; contributed to testing, debugging, **CI/CD**, code reviews, and documentation across Zangoh Studio, Zangoh ASR, AI Avatars, Nexus, and Video Platform.

TECHNICAL SKILLS

- Programming Languages:** C++, Python
- AI/ML:** PyTorch, LangChain, Generative AI, LLMs, Prompt Engineering, AI Agents, ASR
- Backend & Databases:** FastAPI, REST APIs, Pydantic, PostgreSQL, Alembic
- Tools & Platforms:** Docker, Docker Compose, Git, GitHub, CloudEvents, FFmpeg, MediaPipe, Streamlit

PROJECTS

- Graph-Based Neural Cellular Automata for Medical Image Segmentation** [GitHub](#)
Python, PyTorch, GNN, NCA, Medical Imaging *February 2025 – Present*
 - Research Focus:** Designed a hybrid architecture combining Vision GNN (ViG-UNet) and Neural Cellular Automata (Med-NCA) for high-resolution 2D medical image segmentation.
 - Multi-Resolution Refinement:** Integrated multi-resolution NCA for memory-efficient processing and patch-wise refinement of medical images.
 - Performance:** Achieved a Dice score of 0.84 compared to the MedNCA benchmark of 0.81 using a two-stage refinement approach.
- VesselNet: Scleral Vein Segmentation for Health Monitoring** [GitHub](#)
Python, PyTorch, FastAPI, Computer Vision *May 2025*
 - Deep Learning Architecture:** Designed a two-stage model combining U-Net for scleral region isolation and SegNet for precise scleral vein segmentation.
 - Image Processing:** Implemented CLAHE preprocessing and vascular post-processing to enhance visibility and continuity of thin scleral veins.
 - Feature Extraction:** Extracted vein density, tortuosity, width, and branching points from segmented masks for vascular health analysis.
- FieldWise: Agricultural Monitoring Web Application** [Live Demo](#) — [GitHub](#)
Python, Streamlit, Leafmap, Geospatial Analysis *November 2024*
 - Satellite Analysis:** Developed a satellite imagery platform using Sentinel data for crop monitoring, NDVI visualization, and irrigation assessment.
 - Interactive Mapping:** Implemented interactive farm-area visualization using Leafmap for spatial analysis and moisture assessment.
 - AI Integration:** Integrated Gemini API to provide region-specific agricultural insights based on user-selected geographical areas.

ACHIEVEMENTS

- NASA SpaceApp Challenge Hackathon:** Secured 4th position among 40+ teams at PEC Chandigarh
- Astrotape Hackathon:** Achieved 3rd position in intra-college hackathon organized by STAC, IIT Mandi

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Core Team Member, STAC IIT Mandi:** Coordinated hackathon events on National Space Day 2024 and managed technical events under STAC